

Analyzing Different Liquid, Solid and Hybrid Rockets

D. Leung

Oregon State University, Corvallis, Oregon, 97331, United States of America

This paper examines different design concepts for liquid, solid, and hybrid rockets with varying propellant combinations. It will analyze multiple of each propellant combination, to solve for different rocket characteristics, and compare performance to select the most competent rocket design for our requirements, with respect to performance, storage requirement, and environmental impact.

I. Nomenclature

λ	=	Propellant Mass Fraction
Δv	=	Velocity Gain
$\dot{r}_b$	=	Propellant Burning rate
$\dot{r}$	=	Fuel Regression Law
I_{sp}	=	Specific Impulse
O/F	=	Oxidizer to Fuel Ratio
D_t	=	Throat Diameter
D_e	=	Exit Diameter
D_c	=	Chamber Diameter
L_n	=	Nozzle Length
L_c	=	Chamber Length
c^*	=	Characteristic Velocity
$\dot{m}$	=	Mass Flow Rate
r	=	Mixture Ratio
T	=	Thrust
CF	=	Thrust Coefficient

II. Introduction

For this assignment, we are looking to analyze three concepts of booster rockets, using two different propellant types for each. The three concepts are, liquid rocket engine, solid rocket motor, and hybrid rocket engines. For each we will be selecting 2 sets of propellant combinations and calculating the resulting characteristics that our booster must have.

III. Requirements

The booster we are designing is a first stage rocket booster that must provide an ideal velocity gain of $\Delta v = 2300$ m/s (neglecting drag and gravity losses), with a payload mass of 100,000 kg. We are to assume that we have a propellant mass fraction of $\lambda=0.9$. We want our burning time to be less than 100s, and we want the average thrust of our rocket to be roughly twice the booster liftoff mass. Our vehicle diameter is going to be constrained to 3m. For

solid rockets, we are making the assumption that the $\dot{r}_b = 0.4 * Pc^{0.3}$, where Pc is in psi, and $\dot{r}_b$ is in in/s. For the Hybrid rocket, we will assume $\dot{r} = 0.1 * Gox^{0.68}$, where $\dot{r}$ is in in/s, and Gox is in lb/s/in²

For each of the concepts, we will be determining when possible, the optimal O/F ratio, The specific impulse at sea level, the necessary engine/motor and nozzle dimensions, the mass of propellant needed along with the necessary propellant tank size, and the trajectory with time, showing altitude, velocity, and acceleration.

IV. Liquid Engine 1: RP-1 & LOX

For the first liquid engine propellant combination, we will be taking a look at RP-1 (refined kerosene), and liquid oxygen. Using CEA, assuming an initial chamber pressure of 1000 psia, and a pressure ratio of 68, obtained from sea level, we find that the optimum mixture ratio for RP-1 and LOX is going to be 2.3. At a mixture ratio of 2.3, the resulting Isp at sea level from CEA accounting for both equilibrium and frozen flow is calculated to be Isp = 291.2s.

Using our required $\Delta v = Isp * g0 * \ln(m_i/m_f)$, we can then calculate the initial to final mass ratio, which combined with our propellant mass fraction equation $\lambda = m_p/(m_p + m_s)$ leads us to finding the required propellant mass and structural mass. In the case for RP-1 and LOX, the mass of propellant needed was 138397.3kg. With the propellant mass, the structural mass, and the payload mass calculated, we then know the initial takeoff mass, and using that value can find the rough estimate of what we want our average booster mass to achieve. The value found for the minimum average rocket thrust needs to be 5087643.1N. Using the general requested burn time of 100s, the average rocket thrust using the average propellant mass flow rate and Isp is less than the minimum average thrust we need. For that reason, we need to reduce our burn time. To get a higher average thrust, we decrease the burn time to 80s. This results in an average thrust of 5120481N which gives us over our requested average thrust.

Using $At = \dot{c} * \dot{m} / p_0$, we can find the throat area of the nozzle, which ends up being around 0.46m² and the throat diameter is 0.77m. At the optimum O/F ratio, the NASA CEA also gives us a gamma of 1.213, and using the pressure ratio and gamma the area ratio needed is 8.65. Applying that to our calculated throat area by using $Ae = At * area\ ratio$, we get an exit area of 4.06m², and an exit diameter of 2.274m, This predicted exit radius is below the constraining diameter, thus this nozzle size would work. With our area ratio of 8.65, and assuming an 15 degree conical nozzle, our nozzle length to throat diameter ratio is found to be around 7 from the textbook. Our nozzle length ends up being 5.41m, by using $Nozzle\ length = Nozzle\ length\ ratio * Dt$. Taking a concentration ratio of 1.3, and then an L_star value of 2, we calculate a chamber diameter of 0.882m and a chamber length of 1.54m.

Pulling our mass of propellant calculated earlier, we can use the mixture ratio to find the mass of fuel and mass of propellant. We use the equations $m_{fuel} = m_p/(1 + r)$, and $m_{oxidizer} = m_{fuel} * r$. For our O/F ratio of 2.3, we get a resulting mass of 43447.9kg of fuel, and 99930.2kg of oxidizer. For the density of RP-1, I used 811.86 kg/m³, a value found from a journal of propulsion and power. For liquid oxygen, the density of 1141 kg/m³ was found from a cryogenic safety manual from Colorado State University. Using those values we can find the volume needed by dividing the required mass by the density. The fuel volume is 53.517m³, the oxidizer volume is 87.581m³, and overall volume of tanks needed is 141.1m³. Using a concentric tank storage, with the outer diameter of the storage tank being 3m, it is found that the length of the storage tank would be around 19.96m, assuming cylindrical tanks. One of the things not calculated is piping and other inert components to make the rocket work.

Using values collected, we set up an IVP in python to show the altitude, velocity, and acceleration ignoring drag.

For our tank, we are going to assume that the maximum load on the tank is going to occur right at start, as the acceleration combined with the tanks being full will apply the most pressure. Taking our storage tank length, and acceleration of around 10m/s² from our graphed values, we calculate the maximum stress right at takeoff when the tanks are full by using $P = \rho * (g0 + a) * h$. We also split the tank into two tanks, one for fuel and one for

Oxidizer. Using Aluminum 6061 for the tanks, and grabbing numbers from ASM, to get max allowable working stress, we get 220.5MPa. Separating the tanks into fuel and oxidizer, we can then calculate the max forces at the bottom of the tank, and solve for the minimum tank thickness. In this case, for fuel, we would need .83mm, and for oxidizer we would need 1.9mm.

The process and the work for this engine was done in Jupyter Python and is attached in the Appendix.

Liquid Engine 2: LH2 & LOX

For the second liquid engine, we will be using a similar process as the first liquid engine. This time we are looking at liquid hydrogen and liquid oxygen. Using CEA, assuming an initial chamber pressure of 1000 psia, and a pressure ratio of 68, obtained from sea level, we find that the optimum mixture ratio for LH2 and LOX is going to be 3.4. At a mixture ratio of 3.4, the resulting Isp at sea level from CEA accounting for both equilibrium and frozen flow is calculated to be $Isp = 386.8s$.

Using our required $\Delta v = Isp * g_0 * \ln(m_i/m_f)$, we can then calculate the initial to final mass ratio, which combined with our propellant mass fraction equation $\lambda = m_p/(m_p + m_s)$ leads us to finding the required propellant mass and structural mass. In the case for LH2 and LOX, the mass of propellant needed was 91825.33kg. With the propellant mass, the structural mass, and the payload mass calculated, we then know the initial takeoff mass, and using that value can find the rough estimate of what we want our average booster mass to achieve. The value found for the minimum average rocket thrust needs to be 3963792N. Using the general requested burn time of 100s, the average rocket thrust using the average propellant mass flow rate and Isp is less than the minimum average thrust we need. By reducing the burn time to around 87s, we end up with an average thrust produced of 4005325N, which meets our requirement of double the initial take off mass.

Using $At = c^* \cdot \dot{m} / p_0$, we can find the throat area of the nozzle, which ends up being around 0.37m² and the throat diameter is 0.688m. At the optimum O/F ratio, the NASA CEA also gives us a gamma of 1.3008, and using the pressure ratio and gamma the area ratio needed is 7.37. Applying that to our calculated throat area by using $Ae = At * area\ ratio$, we get an exit area of 2.74m², and an exit diameter of 1.868m. This predicted exit radius is below the constraining diameter, thus this nozzle size would work. With our area ratio of 7.37, and assuming an 15 degree conical nozzle, our nozzle length to throat diameter ratio is found to be around 6 from the textbook. Our nozzle length ends up being 4.127m, by using $Nozzle\ length = Nozzle\ length\ ratio * Dt$. Taking a concentration ratio of 1.3, and then an L_star value of 2, we calculate a chamber diameter of 0.784m and a chamber length of 1.54m.

Pulling our mass of propellant calculated earlier, we can use the mixture ratio to find the mass of fuel and mass of propellant. We use the equations $m_{fuel} = m_p/(1 + r)$, and $m_{oxidizer} = m_{fuel} * r$. For our O/F ratio of 3.4, we get a resulting mass of 20869.4kg of fuel, and 70955.9kg of oxidizer. For the density of LH2, I used 70.8 kg/m³, a value found from a paper on hydrogen properties. For liquid oxygen, the density of 1141 kg/m³ was found from a cryogenic safety manual from Colorado State University. Using those values we can find the volume needed. The fuel volume is 294.765m³, the oxidizer volume is 62.2m³, and overall volume of tanks needed is 356.95m³. Taking that volume and our constrained diameter, we can calculate the storage tank sizing the same way as the first liquid engine. We end up having a storage tank length of 50.5m, and a diameter of 3m.

Using values collected, we set up an IVP in python to show the altitude, velocity, and acceleration ignoring drag.

For our tank, we are going to assume that the maximum load on the tank is going to occur right at start, as the acceleration combined with the tanks being full will apply the most pressure. Taking our storage tank length, and acceleration of around 10m/s² from our graphed values, we calculate the maximum stress right at takeoff when the tanks are full. Using Aluminum 6061 for the tanks, and grabbing numbers from ASM, to get max allowable working stress, we get 220.8MPa. Separating the tanks into fuel and oxidizer, we can then calculate the max forces at the

bottom of the tank, and solve for the minimum tank thickness. In this case, for fuel, we would need 0.394mm, and for oxidizer we would need 1.34mm.

The process and the work for this engine was done in Jupyter Python and is attached in the Appendix.

Solid Rocket Motor 1: HTPB + Al + AP

For the first solid rocket motor design, I decided to use HTPB combined with Al, combined with AP (ammonium perchlorate). In this case, from Andoya space, I found that the typical % masses for each of the ingredients is AP around 68%, Aluminum around 20%, and the fuel binder HTPB around 11.5% with some added on curative for the remaining 0.5%. However, I decided to test for myself what the ideal O/F ratio is. In this case, I kept the aluminum to HTPB ratio the same, but tested for different O/F ratios. Based on the best characteristic velocity, the optimal O/F ratio in this case is 2.4. Using these values, in CEA, replacing HTPB with butadiene (C4H6) as it has similar properties, and AP as NH4ClO4(I), along with 1000 psia for initial pressure and exit pressure at sea level, we get a gamma of 1.1301, and a Isp of 286.7s. I tried to look at frozen flow as well, but CEA doesn't output any frozen values past the throat.

Using our required $\Delta v = Isp * g_0 * \ln(m_i/m_f)$, we can then calculate the initial to final mass ratio, which combined with our propellant mass fraction equation $\lambda = m_p/(m_p + m_s)$ leads us to finding the required propellant mass and structural mass. In the case for HTPB + Al + AP, the mass of propellant needed was 147214.7kg. With the propellant mass, the structural mass, and the payload mass calculated, we then know the initial takeoff mass, and using that value can find the rough estimate of what we want our average booster mass to achieve. The value found for the minimum average rocket thrust needs to be 5171280.5N. Using the general requested burn time of 100s, the average rocket thrust using the average propellant mass flow rate and Isp is less than the minimum average thrust we need. By reducing the burn time to around 80s, we end up with an average thrust produced of 5176253N, which meets our requirement of double the initial take off mass.

For the design of our nozzle, we can take the thrust of the rocket, and the CF found at sea level, which was around 1.64, along with our chamber pressure to find our throat area. The formula is $A_t = T/(p_0 * CF)$. This value is found to be $A_t = 0.458m^2$, which translates to 0.76m throat diameter. Taking the area ratio of 10.233 found using the sea level pressure ratio and gamma, we achieve an exit area of $4.686m^2$, and an exit diameter of 2.44m, by $A_e = A_t * area\ ratio$. These values fit our 3m diameter contractions allowing us to move forward. Using an assumed 15 degree conical nozzle, we found that the ratio of length to throat diameter should be around 4, this gets us a nozzle length of 3.05m, by using $Nozzle\ length = Nozzle\ length\ ratio * D_t$.

The mass of propellant also gives us the total volume that is needed to store our propellant. Using values from NIH pubchem, we got the density of aluminum and AP, then from a Japanese scientific journal, we pulled the density of HTPB. Using the densities and the propellant combination ratio, I was able to find the propellant density of around $1977.22kg/m^3$, and thus a volume needed of $74.455m^3$.

Falling within the constraints, we set the outer diameter of the combustion chamber, which acts as the storage for the solid propellant, to 3m. Then, using our new calculated burn time of 80 seconds, we can calculate the supposed web thickness using our propellant burning rate value when the chamber pressure is 1000psia. The equation for this being $\dot{r}_b = 0.4 * P_c^{0.3}$. The propellant burning rate is calculated to be 0.318in/s, and the web thickness needed is 0.646m. Subtracting 2 web thicknesses from the outer diameter, we get an inner diameter of 1.7m. Then, using our volume, we can find what the length of the chamber must be to hold the propellant, and that is calculated out to be 15.6m.

Using values collected, we set up an IVP in python to show the altitude, velocity, and acceleration ignoring drag.

The process and the work for this engine was done in Jupyter Python and is attached in the Appendix.

V. Solid Rocket Motor 2: PVC + Al + AP

For the second solid rocket, I decided to choose a PVC (C_2H_3Cl), aluminum, and ammonium perchlorate fuel. In CEA, testing different fuel percentages, I found that the characteristics velocity caps out when the fuel is split 35% aluminum and 65% PVC, and using that combined with ammonium perchlorate, the optimum O/F ratio is 2. This results in us getting 66% AP, 12% Al, 22% PVC. Using the O/F ratio of 2, we now can get the gamma of 1.2229, and an Isp of 276.55s.

Using our required $\Delta v = Isp * g_0 * \ln(m_i/m_f)$, we can then calculate the initial to final mass ratio, which combined with our propellant mass fraction equation $\lambda = m_p/(m_p + m_s)$ leads us to finding the required propellant mass and structural mass. In the case for PVC + Al + AP, the mass of propellant needed was 156673.6kg. With the propellant mass, the structural mass, and the payload mass calculated, we then know the initial takeoff mass, and using that value can find the rough estimate of what we want our average booster mass to achieve. The value found for the minimum average rocket thrust needs to be 5377484.7N. Using the general requested burn time of 100s, the average rocket thrust using the average propellant mass flow rate and Isp is less than the minimum average thrust we need. By reducing the burn time to around 79s, we end up with an average thrust produced of 5380450N, which meets our requirement of double the initial take off mass.

For the design of our nozzle, we can take the thrust of the rocket, and the CF found at sea level, which was around 1.58, along with our chamber pressure to find our throat area. The formula is $A_t = T/(p_0 * CF)$. This value is found to be $A_t = 0.492m^2$, which translates to 0.79m throat diameter. Taking the area ratio of 8.48 found using the sea level pressure ratio and gamma, we achieve an exit area of $4.17m^2$, and an exit diameter of 2.3m, by $A_e = A_t * area\ ratio$. These values fit our 3m diameter contractions allowing us to move forward. Using an assumed 15 degree conical nozzle, we found that the ratio of length to throat diameter should be around 3.7, this gets us a nozzle length of 2.93m, by using $Nozzle\ length = Nozzle\ length\ ratio * D_t$.

The mass of propellant also gives us the total volume that is needed to store our propellant. Using values from NIH pubchem, we got the density of aluminum and AP, then from the British plastics federation, we pulled the density of PVC. Using the densities and the propellant combination ratio, I was able to find the propellant density of around $2029.8kg/m^3$, and thus a volume needed of $77.19m^3$.

Falling within the constraints, we set the outer diameter of the combustion chamber, which acts as the storage for the solid propellant, to 3m. Then, using our new calculated burn time of 79 seconds, we can calculate the supposed web thickness using our propellant burning rate value when the chamber pressure is 1000psia. The equation for this being $\dot{r}_b = 0.4 * P_c^{0.3}$. The propellant burning rate is calculated to be 0.318in/s, and the web thickness needed is 0.638m. Subtracting 2 web thicknesses from the outer diameter, we get an inner diameter of 1.72m. Then, using our volume, we can find what the length of the chamber must be to hold the propellant, and that is calculated out to be 16.312m.

Using values collected, we set up an IVP in python to show the altitude, velocity, and acceleration ignoring drag.

The process and the work for this engine was done in Jupyter Python and is attached in the Appendix.

VI. Hybrid Rocket Motor 1: HTPB + LOX

For the first propellant combination in hybrid rocket motors, I decided to use LOX combined with HTPB. I used CEA to find the ideal O/F so we can use that as the initial mixture ratio for our design. We know that the O/F ratio will change over time, but we can have the best efficiency right off the bat in order to get off the ground. We end up finding that the ideal O/F ratio is around 2, which gives up a gamma value of 1.2643, and an Isp at sea level of around 289.56s taking into account equilibrium and frozen flow.

Using our required $\Delta v = I_{sp} * g_0 * \ln(m_i/m_f)$, we can then calculate the initial to final mass ratio, which combined with our propellant mass fraction equation $\lambda = m_p/(m_p + m_s)$ leads us to finding the required propellant mass and structural mass. In the case for HTPB + LOX, the mass of propellant needed was 144789.1kg. With the propellant mass, the structural mass, and the payload mass calculated, we then know the initial takeoff mass, and using that value can find the rough estimate of what we want our average booster mass to achieve. Using a burn time of around 80s, we end up getting a produced thrust based off the I_{sp} , that is greater than 2 times the propellant mass. Our produced force on average is going to be 5141007.56N.

With our thrust produced at start, we can then calculate the nozzle sizes needed for our rocket. Using $A_t = T/(p_0 * C_F)$, we end up getting a throat diameter of 0.78m, and an exit diameter of 2.18m, using the area ratio of 7.85 and the equation $A_e = A_t * \text{area ratio}$, calculated using sea level pressure ratio and the found gamma. Then looking at nozzle configurations, we see that for an area ratio of 7.85, you get a nozzle length ratio of around 4. Thus, the resulting nozzle length for a 15 degree conical nozzle is 3.115m. The diameters calculated are smaller than the constraining diameter of 3m, meaning that this nozzle works for our requirements.

Using our initial O/F ratio of 2, we are going to then find our mass flow rate initial of both the fuel and the oxidizer. This gives us a fuel mass flow rate of 606.46kg/s, and an oxidizer mass flow rate of 1212.92kg/s. Then using our constrained diameter as the outer diameter of our chamber, we calculate what the inner diameter must be if the burn time is 80 seconds using the following combustion port radius equation from the textbook:

$$R(t) = \left\{ a(2n+1) \left(\frac{\dot{m}_o}{\pi N} \right)^n t + R_i^{2n+1} \right\}^{1/(2n+1)}$$

This results in an inner diameter of 2.83. This seems like a lot of wasted space, so for that reason, we are going to reduce the size of our chamber's outer diameter to 1.5m. This results in an inner diameter of 0.99m. With that value, we can now calculate the initial fuel regression rate, processing that through, we see that the regression rate is around 0.174 in/s. Using our initial mass fuel flow rate, initial radius, fuel density, and the regression rate, we get a required chamber length of 49.127m. The following equation is how I found chamber length:

$$L = \frac{\dot{m}_f / N}{2\pi R_i \rho_f \dot{r}_i}$$

Using the O/F ratio equation, I imputed my given values and iterated over time. This gave me a resulting O/F ratio as the rocket burned which is located in the tabulated results. The O/F ratio increases over time, and this makes sense because as the port radius increases, the Gox reduces, meaning that less fuel is being burned. If our oxidizer is constant, then the O/F ratio will increase as there will be constant oxidizer and reduced fuel consumption.

With our known oxidizer mass, we can use the oxidizer density to find the oxidizer volume. We use the same density as we have before in the other LOX uses, and get 84.6m³ for the volume. Assuming the outer diameter is going to be at our constrained value of 3m, we calculate the length of the storage tank and it is 11.97m.

Lastly we do a similar analysis as the liquid engine by assuming aluminum 6061, and solving for the required thickness based off of initial takeoff. We see a similar initial acceleration of around 10m/s². We get an allowable working stress of 220.8MPa, assuming an uncrewed mission, and a tank thickness of 1.84mm.

The process and the work for this engine was done in Jupyter Python and is attached in the Appendix.

VII. Hybrid Rocket Motor 2: HTPB + H2O2

For the second propellant combination in hybrid rocket motors, I decided to use H2O2 combined with HTPB. I used CEA to find the ideal O/F so we can use that as the initial mixture ratio for our design. We know that the O/F ratio will change over time, but we can have the best efficiency right off the bat in order to get off the ground. We end up finding that the ideal O/F ratio is around 3.5, which gives up a gamma value of 1.2643, and an I_{sp} at sea level of around 325s taking into account equilibrium and frozen flow. I will note that this seems really

high. I used butadiene as a replacement for HTPB with a chamber temperature of 1000psia, and a pressure ratio of 68 for sea level. Using the ideal efficiency mixture ratio, the Isp values ended up very high.

Using our required $\Delta v = Isp * g_0 * \ln(m_i/m_f)$, we can then calculate the initial to final mass ratio, which combined with our propellant mass fraction equation $\lambda = m_p/(m_p + m_s)$ leads us to finding the required propellant mass and structural mass. In the case for HTPB + H2O2, the mass of propellant needed was 119566.9kg. With the propellant mass, the structural mass, and the payload mass calculated, we then know the initial takeoff mass, and using that value can find the rough estimate of what we want our average booster mass to achieve. Using a burn time of around 80s, we end up getting a produced thrust based off the Isp, that is greater than 2 times the propellant mass. Our produced force on average is going to be 4771093.42N.

With our thrust produced at start, we can then calculate the nozzle sizes needed for our rocket. Using $At = T/(p_0 * CF)$, we end up getting a throat diameter of 0.74m, and an exit diameter of 2.2m, using the area ratio of 8.78 and the equation $Ae = At * area\ ratio$. Then looking at nozzle configurations, we see that for an area ratio of 8.78, you get a nozzle length ratio of around 4.2. Thus, the resulting nozzle length for a 15 degree conical nozzle is 3.122m. The diameters calculated are smaller than the constraining diameter of 3m, meaning that this nozzle works for our requirements.

Using our initial O/F ratio of 3.5, we are going to then find our mass flow rate initial of both the fuel and the oxidizer. This gives us a fuel mass flow rate of 303.18kg/s, and an oxidizer mass flow rate of 1155.63kg/s. Then using our constrained diameter as the outer diameter of our chamber, we calculate what the inner diameter must be if the burn time is 80 seconds using the following combustion port radius equation from the textbook:

$$R(t) = \left\{ a(2n + 1) \left(\frac{\dot{m}_o}{\pi N} \right)^n t + R_i^{2n+1} \right\}^{1/(2n+1)}$$

This results in an inner diameter of 2.84. This seems like a lot of wasted space, so for that reason, we are going to reduce the size of our chamber's outer diameter to 1.5m. This results in an inner diameter of 1.01m. With that value, we can now calculate the initial fuel regression rate, processing that through, we see that the regression rate is around 0.163 in/s. Using our initial mass fuel flow rate, initial radius, fuel density, and the regression rate, we get a required chamber length of 27.84m. This makes sense compared to the previous hybrid motor as with a higher O/F ratio, there is going to be less fuel used comparatively. The following equation is how I found chamber length:

$$L = \frac{\dot{m}_f / N}{2\pi R_i \rho_f \dot{r}_i}$$

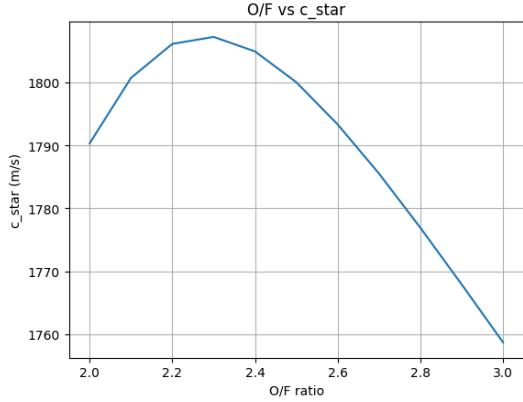
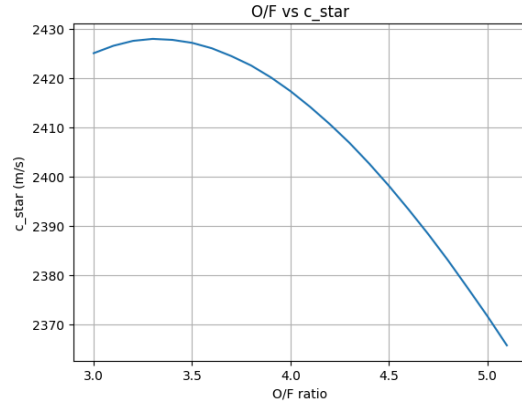
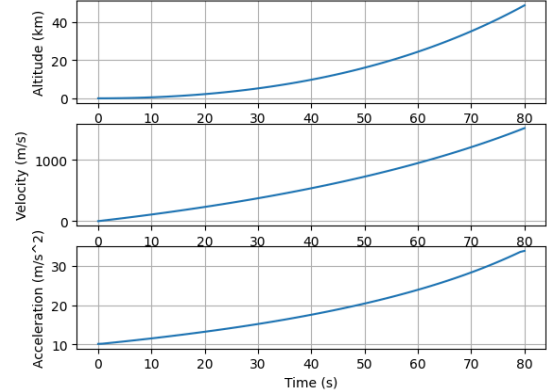
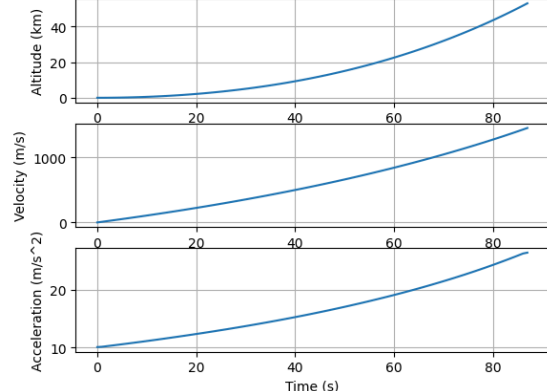
Using the O/F ratio equation, I imputed my given values and iterated over time. This gave me a resulting O/F ratio as the rocket burned which is located in the tabulated results. The O/F ratio increases over time, and this makes sense because as the port radius increases, the Gox reduces, meaning that less fuel is being burned. If our oxidizer is constant, then the O/F ratio will increase as there will be constant oxidizer and reduced fuel consumption

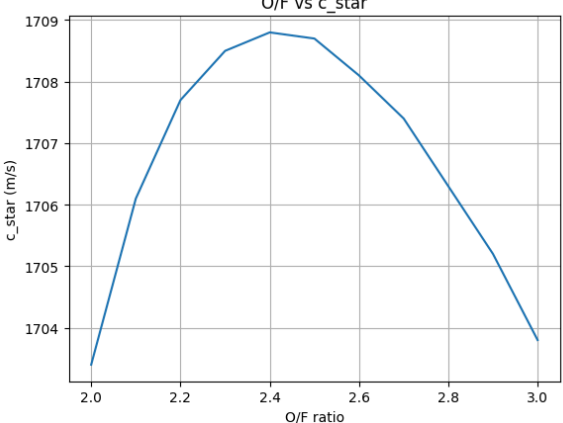
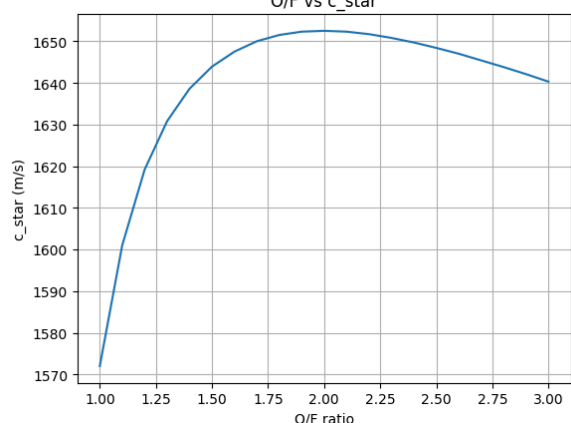
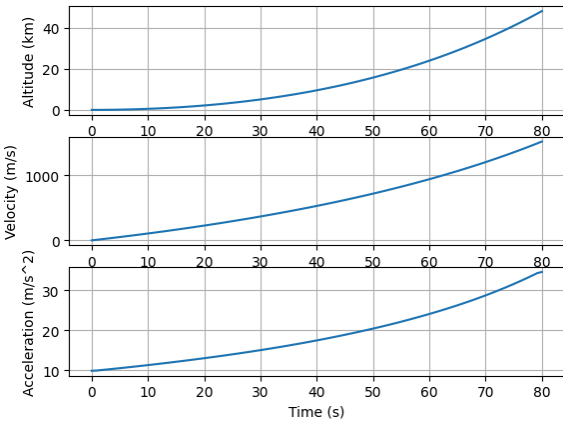
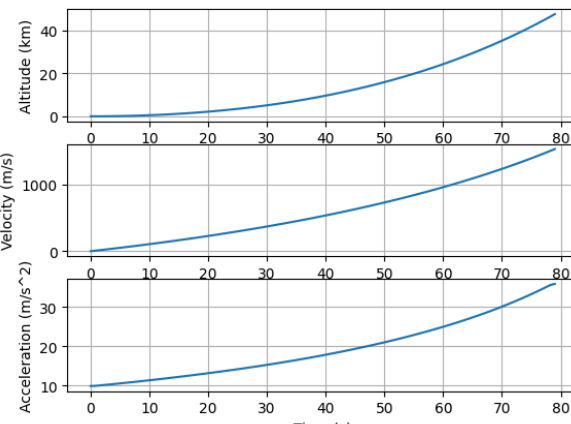
With our known oxidizer mass, we can use the oxidizer density to find the oxidizer volume. We use the same density as we have before in the other LOX uses, and get 63.57m³ for the volume. Assuming the outer diameter is going to be at our constrained value of 3m, we calculate the length of the storage tank and it is 9m.

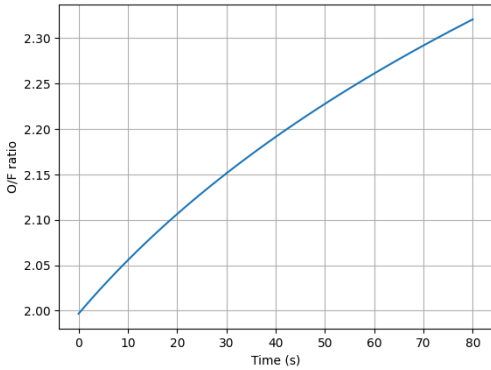
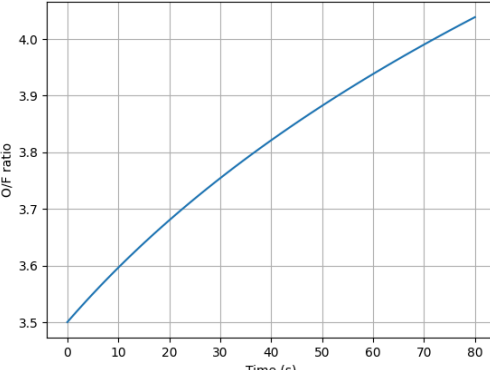
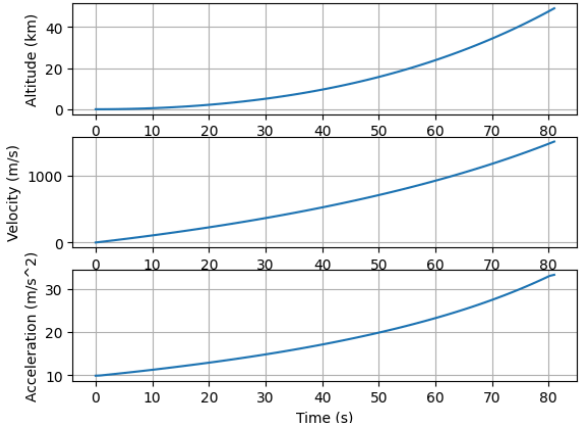
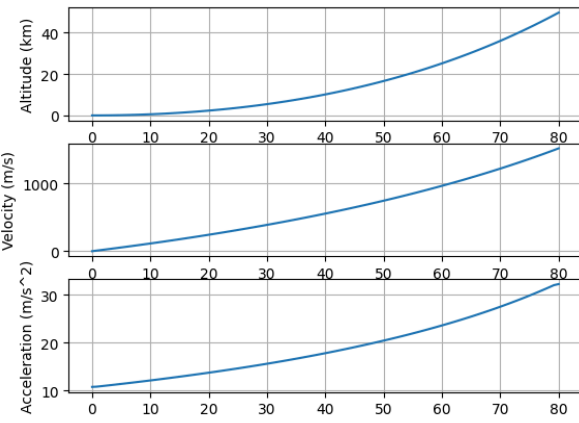
Lastly we do a similar analysis as the liquid engine by assuming aluminum 6061, and solving for the required thickness based off of initial takeoff. We see a similar initial acceleration of around 10m/s². We get an allowable working stress of 220.8MPa, assuming an uncrewed mission, and a tank thickness of 1.77mm.

The process and the work for this engine was done in Jupyter Python and is attached in the Appendix.

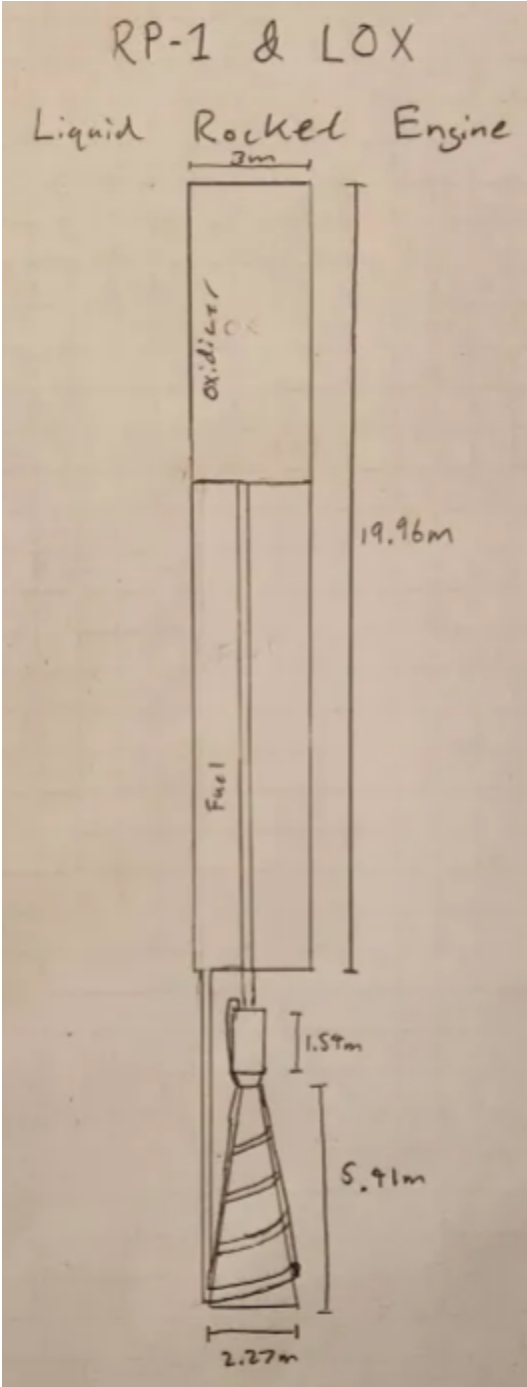
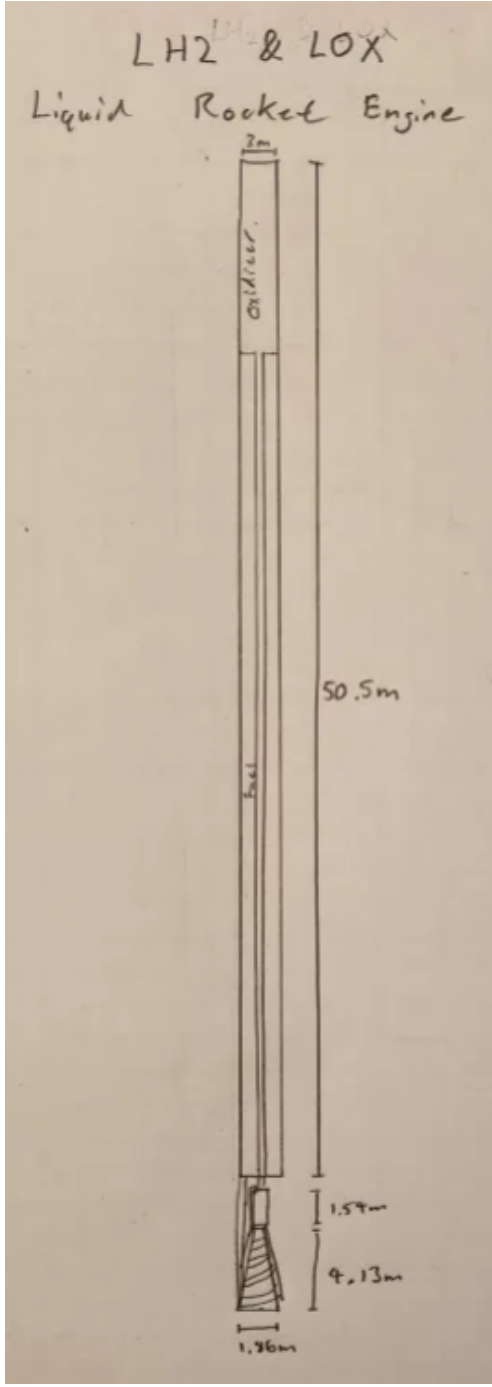
VIII. Tabulated Results and Analysis

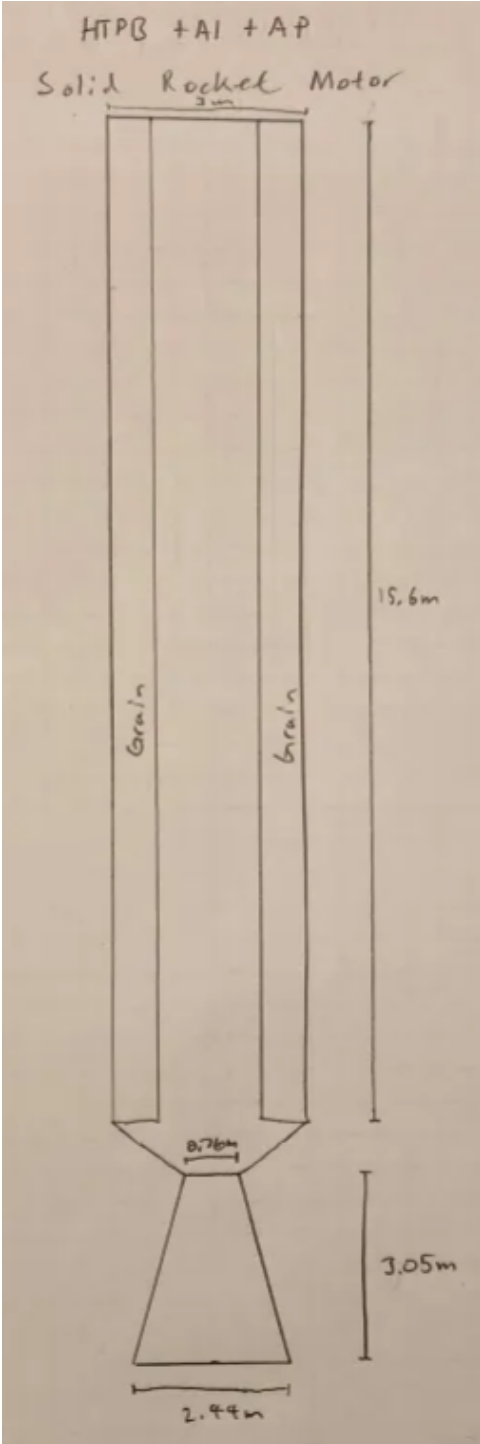
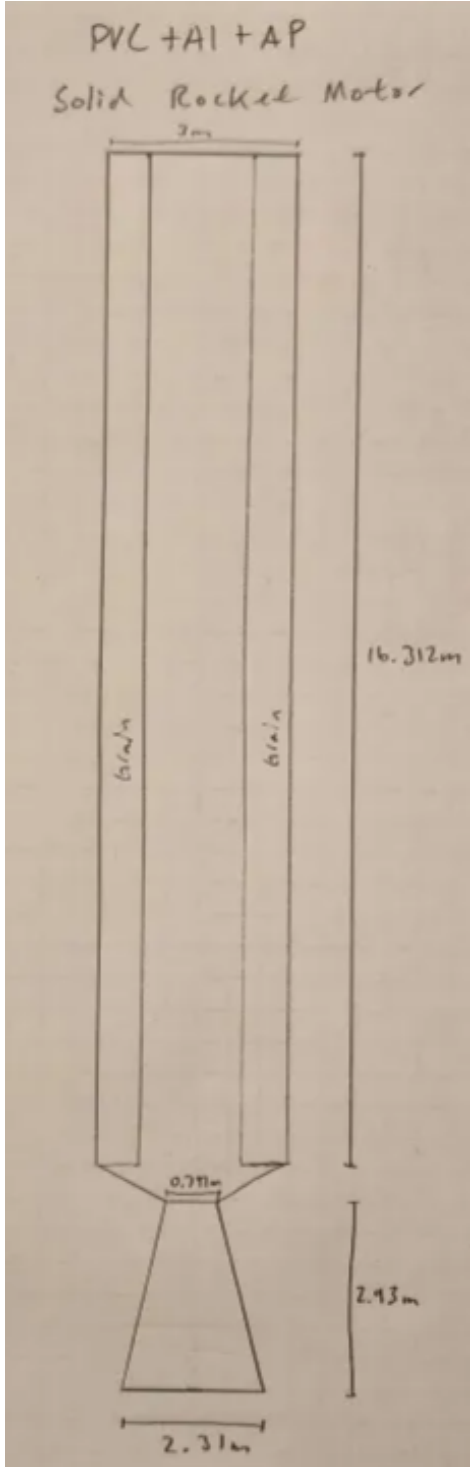
Liquid Engine	
RP-1 & LOX	LH2 & LOX
Optimal O/F = 2.3 	Optimal O/F = 3.4 
Isp = 291.2s	Isp = 387.7
Dt = 0.77m De = 2.27m Dc = 0.882m Ln = 5.41m Lc = 1.54m	Dt = 0.68m De = 1.86m Dc = 0.78m Ln = 4.13m Lc = 1.54m
Propellant Mass = 143378.1kg Propellant Tank Volume = 141.1m³ Tank Length Total = 19.96m Tank Length Fuel = 7.57m Tank Length Oxidizer = 12.39m Tank Diameter = 3m Tank Thickness Fuel = 0.83mm, Aluminum 6061 Tank Thickness Oxidizer = 1.9mm, Aluminum 6061	Propellant Mass = 91825.33kg Propellant Tank Volume = 356.95m³ Tank Length Total = 50.5m Tank Length Fuel = 41.7m Tank Length Oxidizer = 8.8m Tank Diameter = 3m Tank Thickness Fuel = 0.394mm, Aluminum 6061 Tank Thickness Oxidizer = 1.34mm, Aluminum 6061
Rocket Performance for RP-1 & LOX 	Rocket Performance for LH2 & LOX 

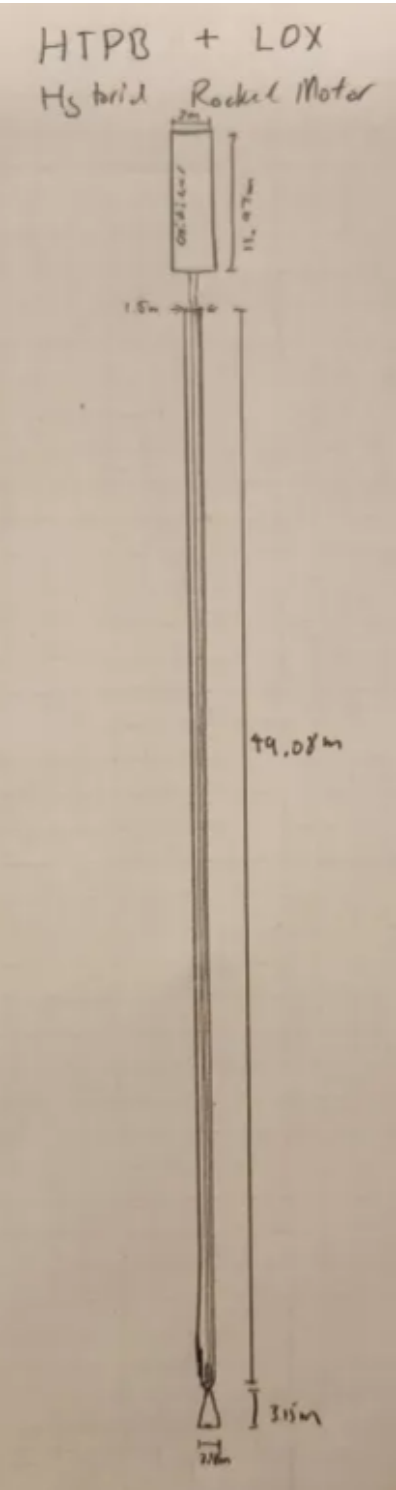
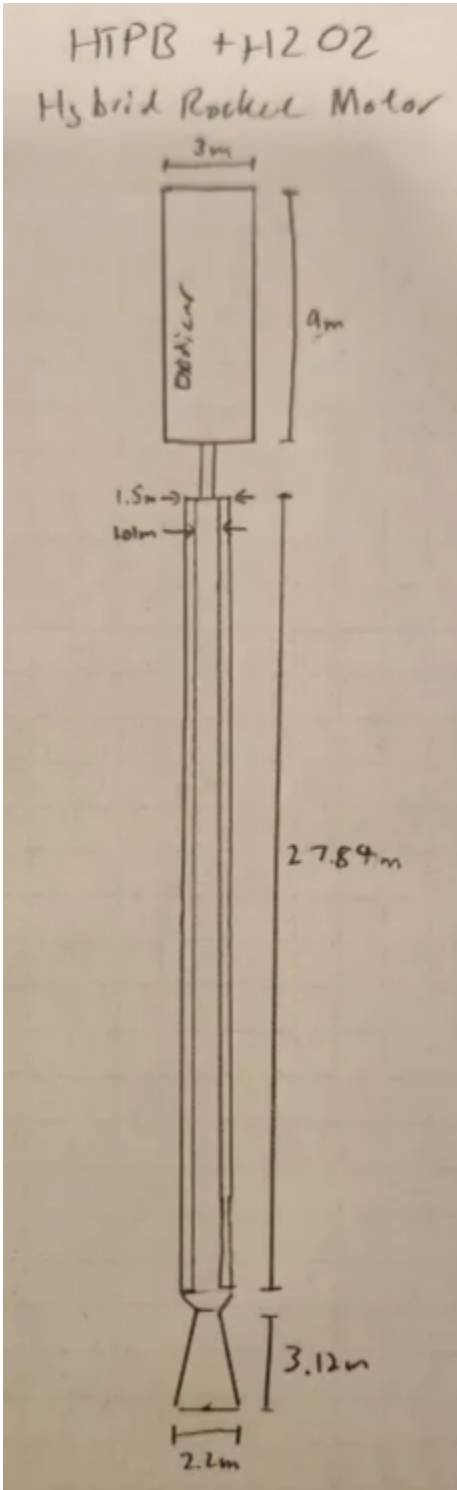
Solid Rocket Motor	
HTPB + Al + AP	PVC + Al + AP
Optimal O/F = 2.4 11% HTPB + 19% Al + 70% AP O/F vs c_{star} 	Optimal O/F = 2 22% PVC + 12% Al + 66% AP O/F vs c_{star} 
Isp = 286.7s	Isp = 276.55s
Dt = 0.76m De = 2.44m Dc = 3m Ln = 3.05m Lc = 15.6m	Dt = 0.792m De = 2.31m Dc = 3m Ln = 2.93m Lc = 16.312m
Propellant Mass = 147214.7kg Fuel Grain Configuration Inner Diameter = 1.71m, Outer Diameter = 3m Chamber Length = 15.6m	Propellant Mass = 156673.61kg Fuel Grain Configuration Inner Diameter = 1.725m, Outer Diameter = 3m Chamber Length = 16.312m
Rocket Performance for HTPB + AL + AP 	Rocket Performance for PVC + AL + AP 

Hybrid Rocket Motor	
HTPB + LOX	HTPB + H2O2
Starting O/F ~ 2 O/F over time 	Optimal O/F ~ 3.5 O/F over time 
Isp = 289.56	Isp = 325.4
Dt = 0.78m De = 2.18m Dc = 1.5m Ln = 3.15m Lc = 49.08m	Dt = 0.74m De = 2.2m Dc = 1.5m Ln = 3.122 Lc = 27.84m
Propellant Mass = 144789.1 Fuel Grain Configuration Inner Diameter = 0.98m, Outer Diameter = 1.5m Chamber Length = 49.08m Oxidizer tank volume = 84.6m³ Oxidizer tank length = 11.97m Oxidizer tank diameter = 3m Tank Thickness Oxidizer = 1.84mm, Aluminum 6061	Propellant Mass = 119566.9 Fuel Grain Configuration Inner Diameter = 1.01m, Outer Diameter = 1.5m Chamber Length = 27.84m Oxidizer tank volume = 63.565m³ Oxidizer tank length = 9m Oxidizer tank diameter = 3m Tank Thickness Oxidizer = 1.77mm, Aluminum 6061
Rocket Performance for HTPB + LOX 	Rocket Performance for HTPB + H2O2 

IX. Rocket Design Sketches

Liquid Engine	
RP-1 & LOX	LH2 & LOX
 RP-1 & LOX Liquid Rocket Engine 2m Oxidizer Fuel 19.96m 1.54m 5.41m 2.27m	 LH2 & LOX Liquid Rocket Engine 2m Oxidizer Fuel 50.5m 1.54m 4.13m 1.36m

Solid Rocket Motor	
HTPB + Al + AP	PVC + Al + AP
	

Hybrid Rocket Motor	
HTPB + LOX	HTPB + H ₂ O ₂
 HTPB + LOX Hybrid Rocket Motor 2m 11.47m 1.5m 19.08m 3.15m 2.2m	 HTPB + H₂O₂ Hybrid Rocket Motor 3m 9m 1.5m 27.84m 3.12m 2.2m

X. Analysis & Discussion of each Rocket Type

When comparing the two liquid rocket engines I designed, the efficiency of LOX & LH2 is significantly greater than that of RP-1 & LOX. However, this efficiency comes at a cost—storage requirements. The total length of the storage tanks increases from 5m for RP-1 & LOX to 12.625m for LOX & LH2, more than doubling the required storage capacity. Another major advantage of RP-1 is its ease of storage; as a refined kerosene, it can be kept at room temperature, whereas LH2 requires cryogenic storage. This makes LH2 much more difficult to handle, transport, and store. One of the biggest advantages of using LOX & LH2 is that its combustion product is primarily water, which has no significant environmental impact. In contrast, RP-1 combustion produces CO₂ emissions, which, while not the worst byproduct of rocket engines, still contribute to environmental concerns. The ease of storage and handling makes RP-1 a strong candidate for first-stage boosters, as has been historically demonstrated. However, I believe that the future of rocket propulsion should prioritize sustainability. While dealing with the storage challenges of LH2 is undoubtedly difficult, the benefits of using a cleaner, more efficient propellant outweigh the downsides. Given these considerations, I would prefer LOX & LH2 over RP-1 & LOX, despite the logistical challenges.

When comparing the two solid rocket motor propellants, HTPB + Al + AP and PVC + Al + AP, the resulting specific impulse (Isp) and overall design configuration are relatively similar. This makes sense, as the primary energetic components—aluminum and ammonium perchlorate—remain unchanged between the two formulations. However, a key distinction lies in their combustion byproducts. While neither option is entirely clean, PVC generates more toxic emissions due to its chlorine content. During combustion, this chlorine tends to form hydrochloric acid (HCl), which can lead to acid rain and cause significant environmental harm. In contrast, HTPB combustion primarily produces carbon dioxide (CO₂) and carbon monoxide (CO), which, while harmful in high concentrations, are less immediately damaging than hydrochloric acid. Excessive CO₂ contributes to global warming, but it is generally considered a less severe pollutant compared to the acid rain potential of PVC combustion. Additionally, PVC does not offer any advantages in terms of storability or efficiency over HTPB. Given its more harmful environmental impact and lack of performance benefits, HTPB + Al + AP is the superior choice for solid rocket propulsion.

When comparing our two hybrid rocket engines—HTPB & LOX versus HTPB & H₂O₂—we observe a considerable difference in their specific impulse (Isp) at sea level. Based on our CEA analysis, while both oxidizers fall within a similar performance range, hydrogen peroxide (H₂O₂) demonstrates greater efficiency. This higher efficiency means that, to achieve the same required thrust, less propellant is needed when using HTPB & H₂O₂. Another key factor is storage. LOX requires cryogenic storage, making it significantly less dense and necessitating a larger oxidizer tank. This is reflected in our design, where the LOX-based rocket requires an oxidizer tank approximately 3 meters longer than its H₂O₂ counterpart. Additionally, the fuel tank for HTPB is smaller in the H₂O₂ design due to a higher initial fuel-to-oxidizer ratio. Evaluating the final sketches, it is evident that the design utilizing H₂O₂ as the oxidizer is superior in terms of efficiency and compactness. The final aspect to consider is environmental impact. Both engines produce similar combustion byproducts: water (H₂O) and carbon dioxide (CO₂) from the HTPB fuel. Since their environmental footprints are nearly identical, and HTPB & H₂O₂ outperforms HTPB & LOX in nearly every other category, the logical choice for our hybrid rocket motor selection is HTPB & H₂O₂.

One noticeable trend across all rocket designs is their similar perceived altitude, velocity, and acceleration profiles. This outcome aligns with our mission requirements, as each rocket was designed to achieve comparable performance metrics. Since the goal was to optimize different propulsion systems while maintaining mission feasibility, it makes sense that all six rockets exhibit similar flight characteristics.

XI. Final Overall Selection

For our final selection, we have narrowed it down to three options: an LH2 & LOX liquid rocket engine, an HTPB + Al + AP solid rocket, and an HTPB & H₂O₂ hybrid rocket. In terms of engine efficiency, the LH2 & LOX combination has a significant advantage over the other two. However, storage presents a major challenge. Due to the low density of liquid hydrogen, an LH2 & LOX rocket requires a much larger volume, making the rocket considerably longer, as shown in the sketch. This size requirement is a significant drawback. Environmental impact is another crucial factor. The LH2 & LOX engine primarily produces water as an exhaust product, which has no

negative environmental impact. In contrast, the HTPB + Al + AP solid rocket produces hydrochloric acid, which can contribute to acid rain, and aluminum oxide, which can be harmful to the atmosphere. Additionally, this solid propellant generates carbon and nitrogen emissions that further impact the environment. The HTPB & H₂O₂ hybrid rocket, while producing water like the LH₂ & LOX engine, also generates some carbon emissions, making it less harmful than the solid rocket but still slightly worse for the environment compared to the liquid rocket. Given the increasing importance of sustainability in rocket propulsion, I believe that despite the challenges associated with storage and size, the LH₂ & LOX liquid rocket engine is the best choice. Its exceptional efficiency and minimal environmental impact outweigh its disadvantages, making it the most suitable option.

Appendix